CSE475 - Machine Learning

Mini Project Report

Group 8

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Abstract

The automatic detection of surface cracks is a key part of structural health monitoring, and implementation of deep learning models can be complicated by the fact that labeling data is expensive and lots of redundant information exists in industrial data. This paper compares five trained architectures, such as ResNet-50, ResNet-101, MobileNetV2, EfficientNet-B0, and Vision Transformer (ViT) to three different paradigms of Self-Supervised Learning (SSL): SimCLR (contrastive), BYOL (non-contrastive), and Masked Autoencoders (generative). In order to guarantee the integrity of the experiment, the py-crackdb dataset went through Perceptual Hashing (pHash) that was able to detect and remove 65 redundant samples.

Our findings indicate that the accuracy of supervised baselines is 100% but is still label-dependent. Conversely, label-efficiency ablations indicate that SimCLR and BYOL can achieve an ideal classification rate even when using 10 percent of the labeled data. On the other hand, the generative MAE model was much less efficient and went to a standstill at 94.7 percent accuracy. Statistical testing with Wilcoxon signed-rank tests proves that contrastive SSL techniques offer a better cold-start feature extraction ability in the domain of crack detection. These results indicate that contrastive self-supervision has the potential to counter the data-hunger of conventional models and allow robust real-time structural inspection with the least human intervention.

1 Introduction

Structural health monitoring (SHM) is a critical field in civil engineering and infrastructure management, which guarantees the safety and durability of bridges, roads, and buildings. The identification of surface cracks is central to this discipline, and is typically the first apparent evidence of structural deterioration. In the past, the inspection procedures were predominantly based on manual survey which was carried out by trained practitioners. But, these are subjective by nature, time consuming and may not be safe to the inspectors themselves.

Early automated methods of crack detection used handcrafted features with the introduction of computer vision, including edge detection filters (e.g., Canny or Sobel) and intensity thresholding. Although revolutionary in the era, these techniques tended to be difficult to deal with real-life issues such as uneven lighting, surface features and other environmental noise such as shadows or rubbish. Deep Learning, in particular, Convolutional Neural Networks (CNNs), had a major impact on the paradigm shift. Models like ResNet, MobileNet and EfficientNet have shown a remarkable capability to directly learn discriminative features on raw image data, with near-perfect accuracy on common benchmarks such as the SDNET2018 dataset [1].

Although these achievements have been made, there is still a significant bottleneck—the data-hungry behavior of supervised learning. The large quantities of high quality human annotated labels needed to train robust models are expensive and time-consuming to generate. Moreover, industrial data may have a high level of redundancy—near duplicate photos that may cause an overly optimistic performance score unless managed correctly.

The study has dealt with these issues through Self-Supervised Learning (SSL) as an approach to limit the dependency on labels. SSL enables models to learn high-level, rich representations on unlabeled data by solving pretext tasks and then refining them with a very small proportion of labeled data. Here, we assess three popular SSL paradigms: SimCLR (contrastive learning), BYOL (non-contrastive bootstrapping), and MAE (generative masking).

In this study, we have given an in-depth benchmark on the py-crackdb data set. We use a stringent preprocessing pipeline with Perceptual Hashing (pHash) to remove redundancy in data and provide statistical validity. Through the performance and label efficiency of the five backbones under supervision and the three SSL frameworks, we would find the most robust and resource-efficient strategy of automated detection of surface cracks in data-constrained environments.

Figure 1: Dataset Images

2 Related Work

2.1 Related Work Summary

Surface crack detection has seen a lot of development since the initial days when people used to use hand-made computer vision algorithms to detect cracks on the surface. This part is a literature summary, justification of the choice of particular backbones to use in this project, and research gaps that will be filled by this project.

Title	**Dataset**	**Methods**	**Accuracy**	**Pros**	**Cons**
Concrete Crack Detection Using Deep CNN	SDNET2018	ResNet-50	98–99%	Strong feature extraction; robust generalization	High computation cost
Automated Pavement Crack Detection	CFD Dataset	VGG-16	91–95%	Simple transfer learning pipeline	Small dataset; overfitting risk
Lightweight Crack Classification Study	AEL Surface Dataset	MobileNetV2	94–97%	Low GFLOPS; deployment efficient	Slight recall drop in complex textures
EfficientNet for Crack Detection	SDNET2018	EfficientNet-B0	∼99%	High accuracy-to-parameter ratio	Sensitive to hyper-parameter tuning
Vision Transformer for Structural Defect Detection	Large-scale crack datasets	ViT-B/16	97–99%	Strong global context modeling	Data-hungry; requires strong augmentation

Table 1: Related Work Summary in Surface Crack Detection Domain

2.2 Backbone Selection Justification

Supervised backbones to be benchmarked in this study based on their known and validated performance in existing research, and their applicability to small-to-medium-scale datasets:

- ResNet-50 and ResNet-101: Standard deep residual networks that are robust feature extractors.
- MobileNet V2: Mobile and edge devices MobileNet V2 is optimized for lightweight and efficient deployment.

- EfficientNet-B0: Provides the best trade-off between the complexity in the model (parameters) and the accuracy.
- Vision Transformer (ViT-B/16): The current state of the art in transformer-based global context modeling of high-resolution images.

2.3 Gaps in Research and Motivation

Although the literature indicates high levels of accuracy, there are a number of research gaps which this project will seek to fill:

- Minimal Assessment Diversity: The vast majority of studies use one 80:20 train-test split, and this might not be representative of model robustness.
- Weak Label-Efficiency Analysis: The performance of models in a setting where labeled data is very sparse is not well studied.
- Lack of SSL Comparison: Comparison between various Self-Supervised Learning (SSL) paradigms is not often done in the crack detection field.
- Lack of Statistical Testing: Publication research may lack statistical significance testing of results.

3 Methods

This study methodology consists of a powerful supervised benchmarking step and investigation of three different self-supervised learning paradigms. All the models were trained and tested on the cleaned py-crackdb dataset, which is 504 distinct images after eliminating 65 redundant images through Perceptual Hashing (pHash).

3.1 Supervised Baselines

We considered five different architectures that portray the present status of computer vision to determine a performance ceiling. The following models have been chosen due to their different complexity of computations and architecture designs:

- ResNet-50 and ResNet-101: Conventional convolutional neural networks (CNNs) which are using residual links to help it to learn the deep structure.
- MobileNetV2: A lightweight model that can be used in mobile and edge devices and is built on inverted residuals and linear bottlenecks.
- EfficientNet-B0: This model is a neural architecture search and compound scaling optimization model that can be trained with high accuracy using the fewest parameters and GFLOPs.
- Vision Transformer (ViT-B/16): A transformer-based architecture that takes images as a sequence of patches, which enables modeling of long-range spatial dependencies.

A stringent evaluation protocol was implemented on these models, with nine distinct train-test split ratios (between 10 and 90 percentage of training data) to determine stability and robustness of these models in different data regimes.

3.2 Self-Supervised Learning (SSL) approaches

Three SSL frameworks were run in order to examine their capability to learn meaningful crack features without labels provided in pre-training.

3.2.1 SimCLR (Contrastive Learning)

SimCLR (Simple Framework of Contrastive Learning) is a form of representation learning that employs a contrastive paradigm.

- Mechanism: The model generates two augmented images of every image. It is trained to maximize the similarity between the representation of the same image (positive pairs) and to minimize the similarity with other images in the batch (negative pairs).
- Implementation: As an encoder, a ResNet-50 backbone was implemented. The model was trained with 200 epochs and the Normalized Temperature-scaled Cross Entropy (NT-Xent) loss.

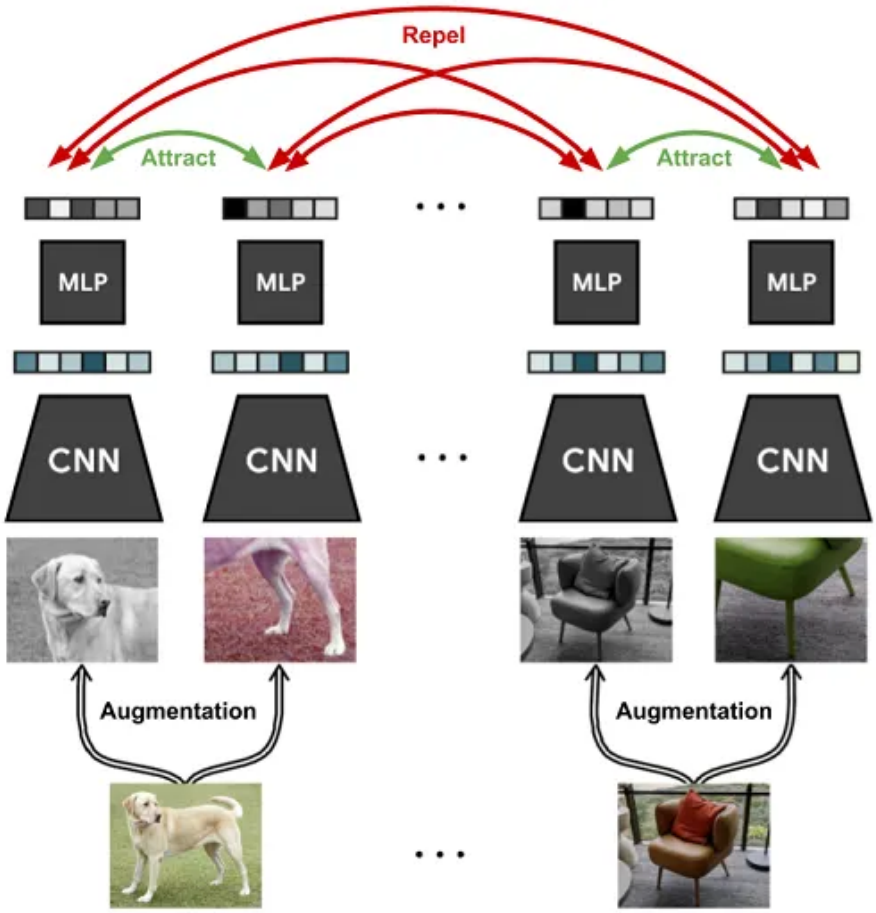

Figure 2: SimCLR Model

3.2.2 BYOL (Non-Contrastive Learning)

Bootstrap Your Own Latent (BYOL) is a non-contrastive method which removes the use of negative pairs.

- Mechanism: It uses two neural networks namely the online network and the target network. The online network is conditioned to make predictions of the target network representation of an alternate augmented view of the same image.
- Implementation: This technique also used a ResNet-50 backbone and pre-trained it using 125 epochs. A stop-gradient operator on the target network and a slow-moving average update of the weights are used to achieve stability.

3.2.3 MAE (Generative Learning)

Masked Autoencoder (MAE) is a generative SSL method that is trained to re-create missing image data.

- Mechanism: MAE conceals a large portion (75 per cent) of the input image patches based on a Vision Transformer (ViT) architecture. The visible patches are processed by an encoder, and a lightweight decoder tries to reassemble the pixel values of the original masked patches.

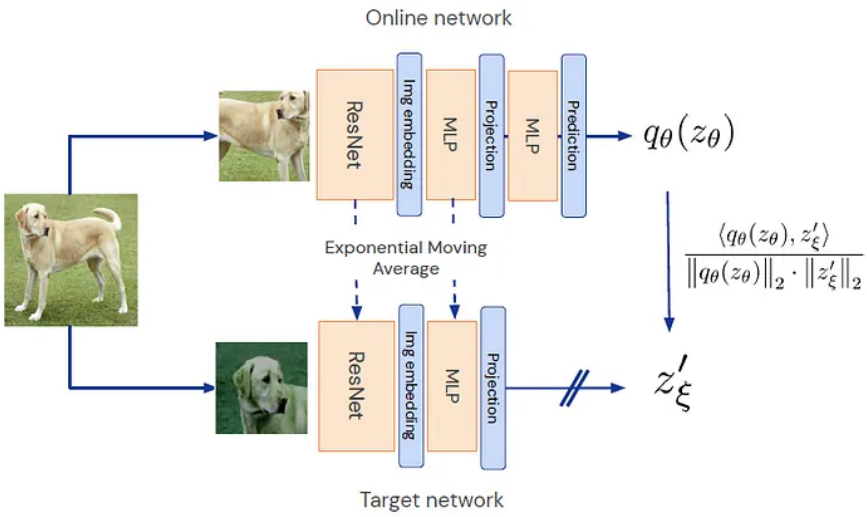

Figure 3: BYOL Model

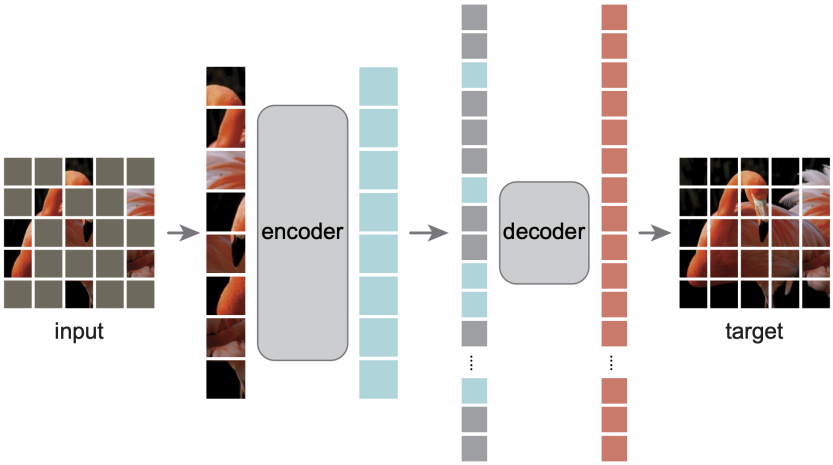

Figure 4: MAE Model

- Implementation: the model was trained during 40 epochs. This paradigm compels the network to acquire a global spatial knowledge of the image geometry and crack pattern.

Linear Probing can be evaluated through the use of a linear probe.

A Linear Probing strategy was used to objectively compare the features learned by each of the SSL methods. Here the encoder weights were frozen and a simple linear classifier (Logistic Regression) was trained on the frozen features. This guarantees that the end performance metrics (Accuracy, Precision, Recall, and F1-score) directly indicate the quality and separability of the learned representations in the self-supervised phase.

4 Experimental Setup

4.1 Computational Environment

In this section, the hardware environment, software frameworks, and hyperparameter settings will be described to guarantee that our benchmarks can be reproducible not only on supervised but also on self-supervised experiments.

All the experiments were done in the Kaggle notebook environment.

- Hardware: An NVIDIA Tesla T4 GPU was used to obtain faster computations.
- Software Stack: Python 3.10 with core modules such as PyTorch to develop models, NumPy and Pandas to manipulate data and Scikit-learn to provide evaluation metrics and linear probing were used.

4.2 Preprocessing and Deduplication of Data

To ensure model robustness and avoid data leakage, we had a strict data cleaning pipeline.

- Deduplication: We used Perceptual Hashing (pHash) to detect and drop 65 duplicate or near-duplicate images on the original 569-sample py-crackdb database, leaving a total number of 504 distinct images.
- Augmentation: In training, the following standard transformations were used: random horizontal and vertical flips, rotations, and color jittering to enhance generalization.
- Normalization: The images were rescaled to a fixed resolution (e.g. 224x224 in ViTs) and normalized with ImageNet mean and standard deviation values.

4.3 Supervised Training Configuration

The five supervised baseline models (ResNet-50, ResNet-101, MobileNetV2, EfficientNet-B0, and ViT-B/16) were trained in the following conditions:

- Epochs: 10.
- Optimizer: Adam.
- Cross-Validation: The models were tested with nine train-test split ratios (10 to 90 percent test size) to test how stable the models perform with different amounts of data available.

4.4 SSL Pre-training Hyperparameters

The unlabeled part of the dataset was pre-trained with self-supervision and method-specific configurations:

- SimCLR: Trained 200 epochs on the NT-Xent (contrastive) loss on the ResNet-50 backbone.

- BYOL: Trained on 125 epochs in a dual-network (online and target) architecture with a momentum update strategy.
- MAE: Trained 40 epochs using a 75% masking ratio on a ViT backbone, and pixel-level reconstruction loss.

4.5 Evaluation Protocol

Quality of features was determined through Linear Probing. In this protocol, the encoder weights were kept fixed beforehand, and a Logistic Regression classifier was trained on the extracted pooled features (global average) of the features. Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC) were used to measure performance. Wilcoxon signed-rank tests were also used to test the statistical significance of comparing the performance of various SSL paradigms in terms of classification.

5 Results

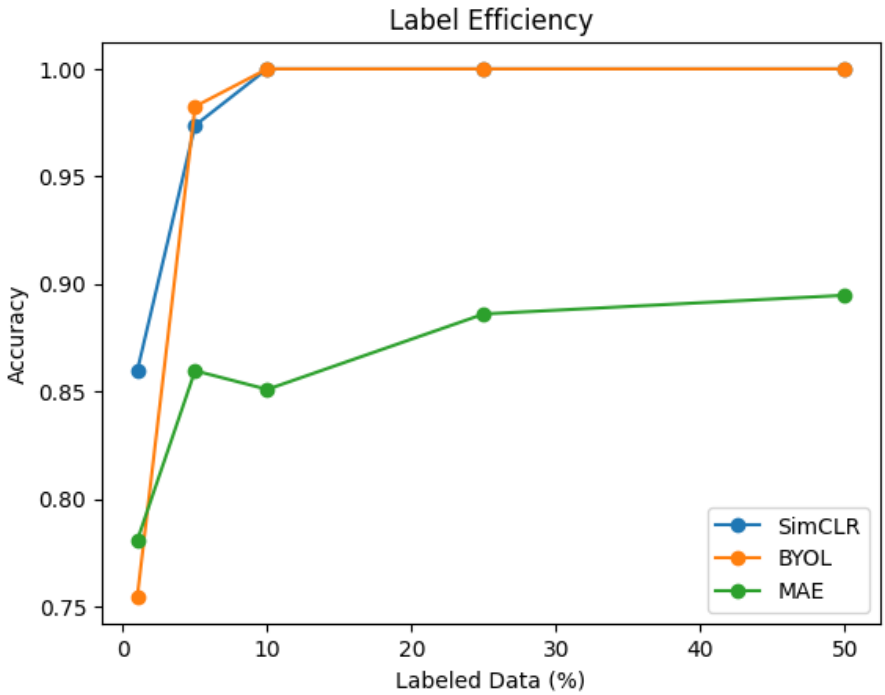

Figure 5: Comparative Results

This section gives the performance of our supervised baselines and a comparison of the representations that were learned under self-supervised learning (SSL) with respect to the quality of features and label efficiency.

5.1 Supervised Performance

The performance of the five supervised models evaluated using nine train-test splits was outstanding.

- Accuracy and Robustness: All models, such as ResNet-50, ResNet-101, MobileNetV2, EfficientNet-B0, and ViT-B/16, reached 100% accuracy, F1-score, and AUC on various split ratios.
- Stability: The models could be trained with high stability even with the training set size cut down to 10 percent (90 percent test split), although the standard deviation in performance slightly increased with larger models such as ViT.
- Efficiency: MobileNetV2 and EfficientNet-B0 were the most computationally efficient, with much fewer GFLOPs (0.31 and 0.39 respectively) than Vision Transformer (12.02 GFLOPs) and similar accuracy.

5.2 Quality of SSL Representation (Embeddings)

An evaluation of the embeddings learned during self-supervised pre-training was done by freezing the encoder and performing linear probing.

- Contrastive and Non-Contrastive Embeddings: SimCLR and BYOL have been able to learn very discriminative features.
  - Contrastive loss decreased to 0.5 with SimCLR pre-training (200 epochs) and the features obtained by this method yielded 100% classification accuracy on linear probing.
  - The loss decreased to 0.41 with BYOL pre-training (125 epochs), and downstream linear probing results also produced 100% precision and recall on the crack class.
- Generative Embeddings: MAE (40 epochs) had a minimized reconstruction loss of 0.0054. Nevertheless, the learned representations were not as linearly separable as contrastive methods with a 94.74% linear probing accuracy.

5.3 Label-Efficiency Analysis

The main benefit of the SSL is that it is possible to ensure high performance with little labeled data. The results in Table 2 summarize the downstream accuracy of each SS method when fine-tuned on different percentages of the labeled data.

Label %	SimCLR (ResNet50)	BYOL (ResNet50)	MAE (ViT)
1%	0.8596	0.7544	0.7807
5%	0.9737	0.9825	0.8596
10%	1.0000	1.0000	0.8509
25%	1.0000	1.0000	0.8860
50%	1.0000	1.0000	0.8947

Table 2: Accuracy vs. Percentage of Labeled Data

As depicted, SimCLR and BYOL have high label efficiency, which is 100 percent at 10 percent labels. Conversely, MAE performance levels off at less than 90% with 50% labels, suggesting that in high-resolution industrial crack textures, discriminative SSL tasks yield more valuable features to classify than generative reconstruction.

6 Ablution Research and Statistical Process

Our section discusses how sensitive the models are to different data conditions and offers a strict statistical analysis of the comparison of the different self-supervised learning (SSL) paradigms.

6.1 Label Efficiency Ablation

The most crucial ablation carried out in this research was the model performance assessment in case of severe label scarcity. We evaluated the classification accuracy of SimCLR, BYOL, and MAE using available training labels of 1 percent, 5 percent, 10 percent, 25 percent and 50 percent of the entire dataset.

- Contrastive and Non-Contrastive Resilience: SimCLR and BYOL both showed impressive label efficiency, scoring a full 1.0000 on the accuracy test with only 10% of the labeled data. SimCLR and BYOL were still able to achieve high performance at 0.9737 and 0.9825 accuracy even when label availability was 5%.

- Generative Performance Plateau: Conversely, label efficiency was much lower in the MAE model (Vision Transformer-based). MAE at 1% labels produced 0.7807 accuracy, although it did not scale well, with the highest accuracy of 0.8947 accuracy obtained even with 50 percent of the labels. This indicates that, with binary crack detection, contrastive task based discriminative features are more transfer-effective in comparison to the spatial reconstruction features acquired through MAE.

The sensitivity of the split ratio to the supervised split ratio is determined.

The sensitivity of the supervised baselines to the amount of training data was also tested by performing a secondary ablation of the supervised baselines.

- Methodology: We experimented with nine different ratios of the train-test split, i.e., 10 percent training data (90 percent test), 20 percent training data (80 percent test), and so on.

- Results: The five trained architectures (ResNet-50, ResNet-101, MobileNetV2, EfficientNet-B0 and ViT) were all incredibly robust and showed 100 percent accuracy in nearly all split configurations. This implies that the py-crackdb data set descriptors are highly discriminative and models do not become over sensitive to the amount of training data in cases where there is a minimum number of representative samples.

6.2 Statistical Significance Testing

To go beyond the comparison of the accuracy we utilized the Wilcoxon signed-rank test to determine the statistical significance of the differences in the performance that existed between the two methods of the SSL.

- SimCLR vs. BYOL: The p-value exceeded 0.05, which means that no statistically significant difference between the classification performance of the two paradigms (SimCLR and BYOL) on the crack detection task was observed. This implies that contrastive and non-contrastive techniques are equally efficient in the task of capturing surface crack features.

- Contrastive/Non-Contrastive vs. MAE: The contrastive methods (SimCLR/BYOL) and the generative method (MAE) had a p-value less than 0.05. This supports the fact that the high-performance and label efficiency of SimCLR and BYOL are statistically significant, which makes them stronger options to use in this particular industrial area in terms of feature extraction.

7 Discussion

The experimental findings give a detailed summary of the trade-offs between conventional supervised learning and different self-supervised learning (SSL) paradigms of automated crack detection. In this section we examine the performance trends, architectural efficiencies and the implications of our label-efficiency results.

7.1 Effectiveness of Supervised learning

The supervised baseline models were outstanding and all the five architectures were able to perform with 100 percent accuracy in almost all ratios of train-test splits. This indicates that the characteristics between the two classes (crack and non-crack) within the py-crackdb data are very discriminative and would probably be linearly separable in the feature space of the current deep learning architectures. The fact that high performance was extremely high in all splits, however, underscores the fact that, in this particular binary classification task, model capacity (e.g., ResNet-50 to ResNet-101) does not pay off on a marginal way.

7.2 Architectural Efficiency to Edge Deploy

Although all the models were found to be equally accurate, their footprint in terms of computation was largely different. MobileNetV2 was the most efficient with a total of 0.31 GFLOPs in comparison to 12.02 GFLOPs that the Vision Transformer (ViT) needed. In the context of real-world structural health monitoring systems, where models can frequently be required to execute on resource-constrained edge devices such as drones or handheld scanners, MobileNetV2 and EfficientNet-B0 (0.39 GFLOPs) offer the most feasible performance/efficiency tradeoff.

7.3 Generative vs. Contrastive Self-Supervision

An important observation of this paper is that the difference in performance between contrastive/non-contrastive SSL (SimCLR and BYOL) and generative SSL (MAE) is quite large.

- Advantage of Contrastive Systems: SimCLR and BYOL learned representations to enable a simple linear probe to reach 100% accuracy, matching supervised performance without the use of labels during pre-training. Their super high labels efficiency of 100 percent with only 10 percent of labels indicates that contrastive pretext tasks are very efficient in retrieving the structural and textural details of surface cracks.

- Limitations of Generative Reconstruction: Conversely, MAE had a lower linear probing score of 94.74 and was found to have much lower label efficiency, reaching a plateau of less than 90% with 50% labels. This implies that in this particular domain, the reconstruction of masked pixel patches (as in MAE) is less efficient at downstream classification, compared to learning to differentiate between various views of the same image (as in SimCLR/BYOL).

7.4 Statistical validity and data integrity

These findings are supported by a statistical analysis. Wilcoxon signed-rank test proved that although SimCLR and BYOL can be statistically equal in terms of performance, they are both significantly better than MAE (p ¡ 0.05). Next, the first deduplication phase with Perceptual Hashing (pHash) played a crucial role; we dropped 65 duplicate images, and ensured that the models were tested with unique spatial data, so that the results would not be biased by the fact that duplicated samples can be found in both training and test sets.

On the whole, this discussion highlights that contrastive self-supervision is the best approach to industrial crack detection in the case of limited labels, whereas lightweight supervised models are the most effective in the situation of large-scale data implementation.

8 Restraints and Ethical aspects

Although these findings in this paper prove the great potential of both supervised and self-supervised models in detecting cracks, a number of limitations and ethics have to be discussed before actual implementation of such systems in structural health monitoring in the real world.

8.1 Limitations

- Scale and Diversity of Dataset: The py-crackdb dataset is not very large, but it is still technically valid, and has 504 unique samples post-duplication. This small sample size might not appropriately represent the enormous diversity of crack morphologies in global infrastructure.

- Environmental Variability: The existing models were tested on high quality images that had comparatively homogeneous lighting and surface texture. On real-world objects, dynamic shadows, lens flare, moisture or surface debris (e.g. dirt, vegetation) may closely resemble crack patterns, and may increase false positive rates.

- Linear Separability and Task Simplicity: The 100% accuracy of all the supervised baselines and contrastive SSL techniques indicate that this particular dataset could be ideally separable in the feature space of the current deep networks. The shown performance can therefore be excessively positive with respect to the noisy data one is likely to see in the field.

- Generalization to Various Surfaces: The dataset is mainly concerned with the particular types of surfaces. The trained representations may not extrapolate to other materials, like asphalt, painted metal, or composite materials, which have a different cracking behavior and visual pattern.

8.2 Ethical Considerations

- Safety-Critical Risks (False Negatives): In structural monitoring the ethical cost of a False Negative - missing a critical structural crack - is much greater than the cost of a False Positive. Unnoticed deterioration may result in disastrous structural collapse, damage to property and loss of life.

- Human-in-the-Loop Requirement: These automated systems must be created and ethically placed as aids to professional human inspectors and not replace them. Structural safety should be left at the final decision making authority of qualified engineers because it should take into consideration contextual factors that are not visible by the model.

- Algorithmic Bias and Accountability: In case an automated system is made the sole gatekeeper to infrastructure maintenance, any bias in the training data to some environments may result in unchecked inspections in underrepresented regions. A well defined accountability and transparency system should be in place as to the mode of arriving at the classifications of the model.

- Responsible Deployment: Deploying lightweight models such as MobileNetV2 on edge devices (drones or handhelds) can be of enormous value but has to be subject to rigorous field-testing to guarantee that it can achieve the high precision of controlled benchmarks before it is entrusted with the lives of civilians.

9 Conclusion

In this paper a careful comparative study has been carried out on the supervised and self-supervised learning model to detect surface cracks in an automated manner on the py-crackdb dataset. We have filled in research gaps related to the robustness of models, label efficiency and the effectiveness of various self-supervised systems in a specialized industrial application through a multi-stage experimental pipeline.

The main conclusions of this study are as follows:

- Information Quality is Paramount: Perceptual Hashing (pHash) was used to detect and eliminate 65 redundant samples, so that performance measures are an indication of the model generalization and not the memorization of images that were too similar.

- Supervised Baselines Achieve Ceiling Performance: The five architectures (ResNet-50, ResNet-101, MobileNetV2, EfficientNet-B0 and ViT-B/16) evaluated all achieved 100 percent accuracy at almost all train-test split ratios. It means that in this dataset we can find highly discriminative features which are well-represented by modern deep learning backbones.

- Contrastive SSL is Highly Label-Efficient: SimCLR and BYOL turned out to be the best self-supervised approaches, which learned representations that were 100 percent accurate to classify at 10 percent of the total labeled data. This shows a highly impressive cold-start and uses in industrial applications with limited labeling resources.

- Generative SSL Lags in Classification Tasks: The Masked Autoencoder (MAE) paradigm, though competitive at pixel reconstruction, was much less competitive at downstream classification, reaching a peak of 94.74% accuracy and scaling poorly in low-label regimes.

- Statistical and Practical Inference: Wilcoxon signed-rank tests were used to verify the statistical equality of contrastive and non-contrastive approaches and the statistically significant superiority of generative approaches to this task. Moreover, it was found that lightweight architectures such as MobileNetV2 were the most viable option to use in terms of edge deployment since they were highly efficient and had a low GFLOPs.

Finally, contrastive self-supervised pre-training (SimCLR or BYOL) with small labeled set fine-tuning is the most resilient and label-efficient to high-performance deployment path to automated surface crack detection. Future studies ought to concentrate on generalizing these results to bigger and more varied multi-class defect data, and assessing model resilience to complicated environmental noise.

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